Koen Essers • Master Internship • April 26, 2026

Notes Week 13

a. Poster

b. Mistake

Even though I looked at the corrigendum from Temmink et al. (2025), and I explicitly set `use_radiation_corrected_transfer_rate` in my models, I very stupidly set it to `.true.` instead of `.false.`. This is fixed from now on.

c. Circumbinary toroid

I looked through MESA's source code for its implementation of the circumbinary coplanar toroid. It is based on the same 2 parameters as Soberman et al. (1997), where δ determines how much of the *transferred* mass is lost to a toroid. Specifically, they define `mdot_system_cct = b% mtransfer_rate * b% mass_transfer_delta`. This is set as follows:

```
! $MESA_DIR/binary/private/binary_private_def.f90
! subroutine adjust_mdots

! mdot_system_transfer is mass lost from the vicinity of
 ↪  each star
! due to inefficient rlof mass transfer, mdot_system_cct
 ↪  is mass lost
! from a circumbinary coplanar toroid.
if (b% mtransfer_rate+b% mdot_wind_transfer(b% d_i) >= 0
 ↪  .or. b% CE_flag) then
    b% mdot_system_transfer(b% d_i) = 0d0
    b% mdot_system_transfer(b% a_i) = 0d0
    b% mdot_system_cct = 0d0
else
    b% mdot_system_transfer(b% d_i) = b% mtransfer_rate *
     ↪  b% mass_transfer_alpha
    b% mdot_system_cct = b% mtransfer_rate * b%
     ↪  mass_transfer_delta
    if (b% point_mass_i == 0 .or. b% model_twins_flag)
     ↪  then
        b% mdot_system_transfer(b% a_i) = b%
         ↪  mtransfer_rate * b% mass_transfer_beta
    else
        ! do not compute mass lost from the vicinity
         ↪  using just mass_transfer_beta, as
        ! mass transfer can be stopped also by going past
         ↪  the Eddington limit
        b% mdot_system_transfer(b% a_i) =
         ↪  (actual_mtransfer_rate + b% component_mdot(b%
         ↪  a_i)) &
            + b% mtransfer_rate * b% mass_transfer_beta
    end if
end if
```

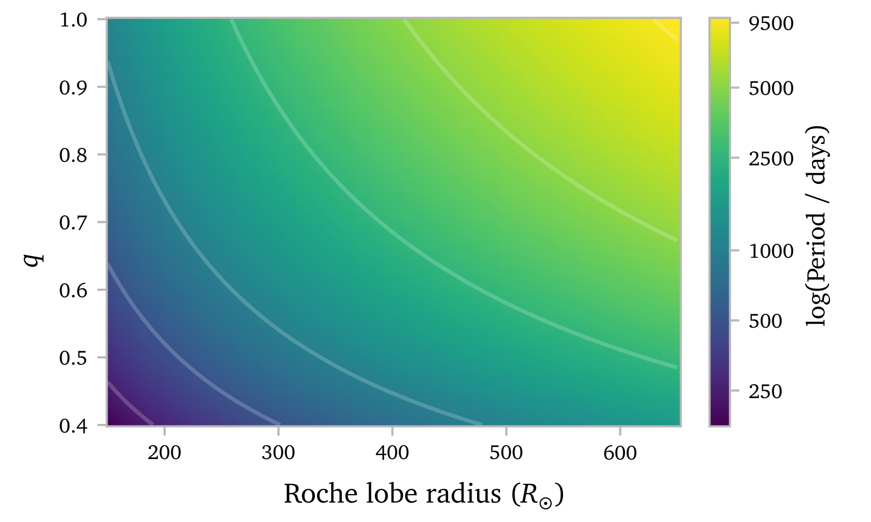
This is then used to compute the change in angular momentum in the default routine as:

```
! $MESA_DIR/binary/private/binary_jdot.f90
! subroutine default_jdot_ml

!mass lost from circumbinary coplanar toroid
b% jdot_ml = b% jdot_ml + b% mdot_system_cct * b% mass_transfer_gamma * &
     sqrt(standard_cgrav * (b% m(1) + b% m(2)) * b% separation)
```

I think just copying this line into my `wind_loss` subroutine should be all that is necessary to simulate mass loss from a coplanar circumbinary toroid. The new `wind_loss` subroutine is now:

```
! $RUN_DIR/src/bin/additional_routines.inc

subroutine wind_loss(binary_id, ierr)
    integer, intent(in) :: binary_id
    integer, intent(out) :: ierr
    type(binary_info), pointer :: b
    real(dp) :: q, vw_over_vorb
    real(dp) :: beta, eta
    real(dp) :: omega, I_eff
    real(dp) :: domega_dt_wind

    ierr = 0
    call binary_ptr(binary_id, b, ierr)
    if (ierr /= 0) then
        write (*, *) 'failed in binary_ptr'
        return
    end if

    ! --- compute q == m_accretor / m_donor and v
    q = b%m(b%a_i)/b%m(b%d_i)
    vw_over_vorb =
     ↪  compute_vwind(b%s_donor)/compute_vorbit(b%m(1) +
     ↪  b%m(2), b%separation)
    b%s_donor%xtra(x_vw_over_vorb) = vw_over_vorb

    ! --- get beta and eta from Saladino
    beta = compute_beta_Sal(q, vw_over_vorb)/sqrt(1 -
     ↪  pow2(b%eccentricity))
    b%s_donor%xtra(x_beta) = beta
    eta = compute_eta_Sal(q, vw_over_vorb)
    b%s_donor%xtra(x_eta) = eta

    ! --- star angular momentum loss due to winds
    ! compute the change in spin frequency. this
    ! gets used in extra_binary_finish_step.
    omega = b%s_donor%xtra(x_omega)
    I_eff = b%s_donor%xtra(x_I_eff)
    domega_dt_wind = (b%mdot_system_wind(b%d_i) / (1 -
     ↪  beta)
     ↪  *(b%s_donor%photosphere_r*Rsun)**2*omega/1.5)/I_eff
    b%s_donor%xtra(x_domega_dt_wind) = domega_dt_wind

    ! --- mass lost from vicinity of donor
    b%jdot_ml = b%mdot_system_wind(b%d_i) &
                *eta &
                *(b%separation)**2 &
                *sqrt(1 - pow2(b%eccentricity)) &
                *2*pi/b%period

    ! --- mass lost from circumbinary coplanar toroid
    b% jdot_ml = b% jdot_ml + b% mdot_system_cct * b%
     ↪  mass_transfer_gamma * &
                 sqrt(standard_cgrav * (b% m(1) + b%
                  ↪  m(2)) * b% separation)

end subroutine wind_loss
```

I simulate an analytic grid again using the same code I used last week to find a nice balance for predicted orbital periods on the MESA grid parameter space. I find that $\delta = 0.2$, $\gamma = 1.25$ gives the following set of predicted orbits:



These values give a very broad region in the parameter space that generates periods between 1000 and 5000 days, with a few smaller and larger periods. I think this aligns well with the observed period distribution of barium stars as well, and hopefully should not be *too* challenging for mesa, since we are not trying to create extremely short periods.

Temmink et al. (2023) Escorza & De Rosa (2023) Jorissen et al. (2019) Rees et al. (2024) Preece et al. (2022) Saladino et al. (2018) Kolb & Ritter (1990)

References

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Jorissen, A., Boffin, H. M. J., Karinkuzhi, D., et al. (2019) *Barium and related stars, and their white-dwarf companions. I. Giant stars*. A&A626, A127.

Kolb, U. & Ritter, H. (1990) *A comparative study of the evolution of a close binary using a standard and an improved technique for computing mass transfer.*. A&A236, 385.

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